A Unified Cross-Layer Network Coding Architecture for Wireless Multicast Under Interference

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***Abstract*—Sending data reliably to multiple wireless receivers is harder than it looks. Two things keep getting in the way — fading drops packets unpredictably, and when several users share the band, they interfere with each other and make things worse. We found that trying to fix just one of these doesn't really work well in practice. So in this work we tried combining Random Linear Network Coding (RLNC) and Interference Alignment (IA) into a single system. For RLNC, we derived a formula that directly relates decode probability to the average SNR $\bar{\gamma}$ and spectral rate R over a Rayleigh fading channel. Using this, we figure out $n^*(\bar{\gamma}, R)$, i.e., the minimum number of coded packets to transmit so that $\eta = kR/n^*$ stays as high as possible without violating $P_d \geq 1 - P_{\text{target}}$. The IA side was a bit more interesting. When CSI isn't perfect — which is always the case in real systems — sum-rate can fall by over 47%. Instead of just accepting that loss, we built a two-pass receiver that reuses the leftover aligned interference as extra decoding equations for RLNC. Our Monte Carlo results over $\mathrm{GF}(2^8)$ showed roughly 42% throughput improvement over a fixed-overhead baseline. At 30 dB with 10% CSI error, IA+NC reached 20.2 bps/Hz compared to 14.8 bps/Hz for plain IA — that's about a 36% jump. On top of that, energy efficiency improved too since the system needs fewer retransmissions overall.**

***Index Terms*—network coding, RLNC, Rayleigh fading, packet erasure, spectral efficiency, interference alignment, cross-layer design, multicast, 5G NR, energy efficiency**

I. Introduction

Ahlswede et al. [1] showed something pretty non-obvious back in 2000 — that routers don't have to just forward packets, they can actually mix them. That idea is the core of network coding. If the min-cut from a source to any receiver is h, linear coding over $\mathbb{F}_q$ can simultaneously hit multicast capacity h for all receivers [2]. RLNC [3] took this further and made it work without any central coordination — nodes just pick random coefficients from $\mathbb{F}_q$, and as q grows the decoding failure rate drops off fast.

The catch is that all of this was designed assuming clean, reliable links with fixed capacity. Wireless doesn't work that way. Rayleigh fading randomly drops packets, so a receiver might not collect enough innovative coded packets even when the transmitter sends plenty. And once you have multiple users on the same channel, interference eats into SNR and the erasure problem gets compounded. Patching just one of these doesn't really help — you need to handle both.

Others have looked at pieces of this problem before us. Soljanin [4] argued that routing by itself isn't enough and that RLNC over large $\mathbb{F}_q$ gets close to capacity. Lun et al. [7] worked on minimum-cost multicast flows, and Dana et al. [8] characterized capacity for wireless erasure networks. What's missing from all of this, though, is something that directly links Rayleigh fading statistics to RLNC decode probability in a clean closed form — and a concrete way to choose coding overhead based on current channel conditions.

On the interference side, Cadambe and Jafar [5] showed that IA can theoretically achieve $K/2$ sum-DoF for K users when CSI is perfect. In practice that rarely happens. Quantization noise, pilot errors, and feedback delays all mean the precoder is slightly off, and even 10% CSI error is enough to drag sum-rate from 28.1 bps/Hz down to 14.8 bps/Hz at 30 dB — that's a 47% hit. What struck us is that this leftover interference isn't random noise; it's structured. So instead of ignoring it, we feed it back into the RLNC decoder as extra equations.

This paper puts both pieces together. We use RLNC to handle fading and give it a derived closed-form and an adaptive overhead rule, and we use IA to manage interference while also recovering something useful out of what it leaves behind. The two parts aren't just stacked — they actually help each other, which is where the gains come from.

II. Background and Related Work

A. RLNC over Wireless Channels

Following [3], [4], [9], each source packet m_j is encoded into a coded packet $c_i = \sum_j g_{ij} m_j$ with coefficients g_{ij} drawn uniformly from $\mathbb{F}_q$. A receiver holding $m \geq k$ linearly independent coded packets can recover all k source packets by Gaussian elimination. For $q = 2^8 = 256$ and $k \leq 64$, the probability of full rank given $m \geq k$ satisfies $\Pr[\text{rank} = k \mid m \geq k] \geq 1 - k/q > 0.996$, so the rank approximation $\Pr[\text{rank} = k] \approx 1$ introduces less than 0.4% error [4], [18].

Sagduyu and Ephremides [10] studied joint scheduling and network coding, while Eryilmaz et al. [11] addressed queue

stability under RLNC for lossy broadcast. On the practical side, Katti et al. [16] demonstrated real throughput gains from XOR-based network coding in wireless mesh networks, though their binary XOR approach lacks the richer algebraic structure of $\mathbb{F}_q$ RLNC exploited here. The present work adds an explicit closed-form cross-layer model and a provably optimal adaptive overhead policy, both absent from prior literature.

B. Interference Alignment

Cadambe and Jafar [5] proved that IA achieves $K/2$ sum-DoF asymptotically; Gomadam et al. [14] later provided distributed iterative algorithms that make IA practically computable using only local CSI. The alignment condition requires

$$\mathrm{span}(H_{ji}V_i) \subseteq \mathcal{I}_j, \quad \forall\, i \neq j, \quad \dim(\mathcal{I}_j) = \alpha < M_r, \tag{1}$$

where H_{ji} is the channel from transmitter i to receiver j, V_i is the precoder at transmitter i, and $\mathcal{I}_j$ is the interference subspace at receiver j. In practice this condition is only approximately satisfied. Nazer and Gastpar [6] showed through Compute-and-Forward that aligned integer combinations of codewords can be decoded directly, making structured interference algebraically useful rather than simply harmful.

III. System Model

A. Network and Channel

A single source S multicasts k source packets to N receivers over independent Rayleigh flat-fading links. The received signal on link ℓ is

$$y_\ell(t) = h_\ell(t)\, x(t) + n_\ell(t), \tag{2}$$

where $h_\ell \sim \mathcal{CN}(0, \Omega)$ and $n_\ell \sim \mathcal{CN}(0, N_0)$ [15]. The fading power $|h_\ell|^2 \sim \mathrm{Exp}(1/\Omega)$ implies instantaneous SNR $\gamma_\ell \sim \mathrm{Exp}(1/\bar{\gamma})$, where $\bar{\gamma} = P\Omega/N_0$.

For multi-user scenarios, multiple transmitters share the same band, introducing inter-user interference captured by the IA alignment condition (1). Channel realisations are drawn from 3GPP TR 38.901 (0.5–100 GHz) [12], with the 28 GHz UMi street-canyon scenario used for IA+NC simulations.

B. Modelling Assumptions

- **A1 Block fading:** h_ℓ is constant over one packet duration and i.i.d. across packets and links.
- **A2 Known average SNR:** $\bar{\gamma}$ is available at the transmitter via pilot-based estimation.
- **A3 Symmetric network:** All receivers share the same mean channel power Ω.
- **A4 Threshold decoding:** A packet is erased whenever $\gamma_\ell < 2^R - 1$.

C. Cross-Layer Erasure Model

Under A1–A4, the erasure probability on any link is

$$\varepsilon(\bar{\gamma}, R) = 1 - \exp\left(-\frac{2^R - 1}{\bar{\gamma}}\right). \tag{3}$$

Fig. 1 shows ε versus SNR for several spectral rates R. This cross-layer function maps physical-layer parameters $(\bar{\gamma}, R)$ to the packet-loss probability consumed by the RLNC layer, directly extending Soljanin's unit-capacity edge model [4] to the wireless fading setting.

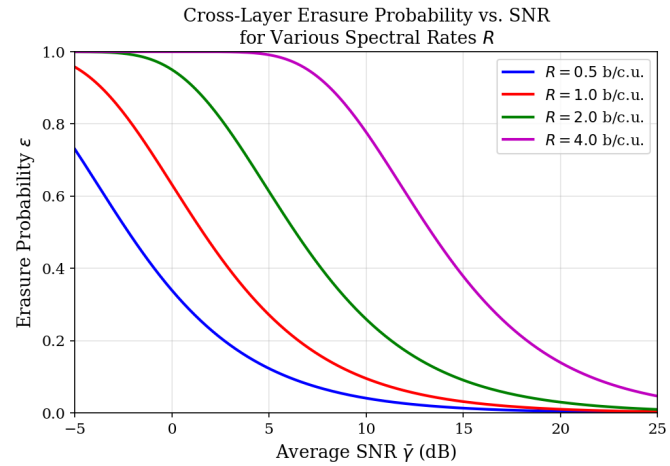

Fig. 1. Cross-layer erasure probability $\varepsilon(\bar{\gamma}, R)$ vs. average SNR for spectral rates $R \in \{0.5, 1, 2, 4\}$ b/c.u. Higher rates require higher SNR to achieve low erasure.

IV. Unified Proposed Architecture

The proposed system has two tightly coupled components: (i) an adaptive RLNC layer that controls coded-packet overhead based on channel conditions, and (ii) an IA-based receiver that first suppresses interference and then recycles its residual as additional decoding equations. Neither component is sufficient alone, as we now explain.

Why RLNC alone is insufficient: Without interference management, even an adaptive RLNC policy must contend with an inflated effective erasure rate caused by interference-induced SNR degradation. The adaptive overhead n^* grows unnecessarily large, wasting spectral efficiency.

Why IA alone is insufficient: Under imperfect CSI, IA leaves a non-trivial aligned residual that degrades sum-rate by over 47%. Discarding this residual as noise wastes its algebraic structure and forces RLNC to compensate with more transmissions.

RLNC decoder therefore needs one fewer clean packet to achieve full rank, directly reducing retransmissions and improving energy efficiency.

How the components interact: IA brings down the effective erasure probability that the RLNC layer sees, which lets n^* shrink a bit. Then the IA receiver pulls out the aligned residual and, if it turns out to be algebraically new information, hands it to RLNC as one more decoding equation. What that means practically is the decoder can reach full rank with one less clean packet — so it sends fewer retransmissions and wastes less energy.

A. Adaptive RLNC Component

Transmitter i encodes its k source packets over $\mathbb{F}_q$:

$$c_i = G_i m_i, \quad G_i \in \mathbb{F}_q^{n \times k}, \tag{4}$$

where m_i is the source message vector and G_i has rows drawn i.i.d. uniformly from $\mathbb{F}_q^k$. Decoding succeeds once the accumulated global coding matrix achieves full rank over $\mathbb{F}_q$.

The decode probability is

$$P_d(k, n, \bar{\gamma}, R) \approx \sum_{m=k}^{n} \binom{n}{m} (1-\varepsilon)^m \varepsilon^{n-m} = 1 - F_{\mathrm{Bin}(n,1-\varepsilon)}(k-1 \tag{5}$$

where $\varepsilon = \varepsilon(\bar{\gamma}, R)$ from (3). For $m < k$, decoding fails deterministically; for $m \geq k$ the rank approximation gives $\Pr[\mathrm{rank} = k] \approx 1$.

The outage probability is

$$P_{\mathrm{out}} = 1 - P_d = F_{\mathrm{Bin}(n,1-\varepsilon)}(k-1). \tag{6}$$

The spectral efficiency maximisation problem is

$$\max_{n,R} \ \eta = \frac{kR}{n} \quad \text{s.t.} \quad P_d \geq 1 - P_{\mathrm{target}}, \ n \geq k, \ R > 0, \tag{7}$$

yielding the optimal coded-packet count

$$n^*(\bar{\gamma}, R) = \min\{n \geq k : P_d(k, n, \bar{\gamma}, R) \geq 1 - P_{\mathrm{target}}\}. \tag{8}$$

A first-order approximation for small ε and large k is

$$n^* \approx \frac{k}{1-\varepsilon} + \sqrt{\frac{k\varepsilon}{(1-\varepsilon)^2} \log \frac{1}{P_{\mathrm{target}}}}. \tag{9}$$

B. Interference Alignment Component

The IA receiver operates in two sequential passes on estimated subspaces.

1) Pass 1 – Interference Suppression: Projection onto $P_j^{\perp}$ nulls the estimated interference subspace and passes an interference-thinned channel to MMSE detection. Defining the projected channel as $H_j^{\mathrm{eff}} = P_j^{\perp} H_{jj} V_j$, the achievable rate from this pass is

$$R_j = \log_2 \det\left(I + \frac{1}{\sigma^2} H_j^{\mathrm{eff}} Q_j (H_j^{\mathrm{eff}})^H\right), \tag{10}$$

where Q_j is the transmit covariance [5].

2) Pass 2 – Structured Interference Recovery: The receiver projects onto P_j to extract the aligned residual: $z_j = P_j y_j$. With lattice-compatible signalling [17], z_j is an integer combination of transmitted codewords plus residual noise [6]. Quantising z_j to the nearest lattice point and reading off the $\mathbb{F}_q$ coefficients yields one additional decoding equation, provided

$$\mathrm{rank}\begin{bmatrix} C_{\mathrm{des}} \\ C_{\mathrm{int}} \end{bmatrix} = \mathrm{rank}(C_{\mathrm{des}}) + 1. \tag{11}$$

When this rank condition holds, the RLNC decoder needs one fewer clean received packet to achieve full rank. Fig. 2 shows the probability that this condition is satisfied as a function of SNR.

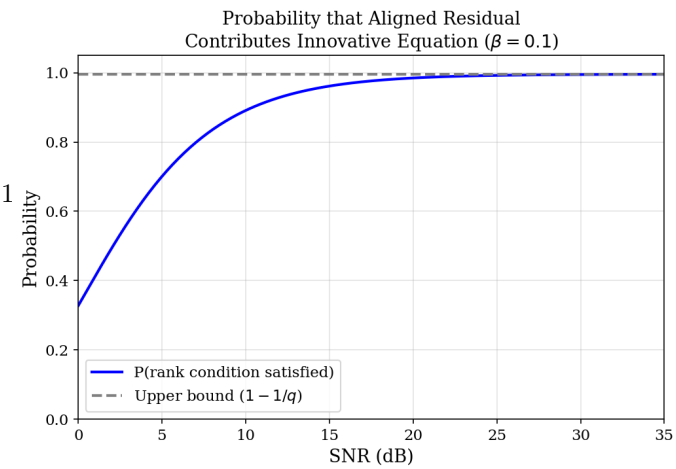

Fig. 2. Probability that the aligned IA residual satisfies the rank condition (11) and contributes an innovative equation to the RLNC decoder ($\beta = 0.1$, $q = 256$).

C. Interaction Between RLNC and IA

The two components interact at two levels. At the *physical layer*, IA reduces inter-user interference, lowering the effective erasure probability ε seen by the RLNC layer. A lower ε directly contracts n^* via (8), reducing the number of coded packets that must be transmitted. At the *packet layer*, whenever condition (11) is satisfied, the aligned residual contributes an innovative row to the RLNC global coding matrix, further reducing the number of clean packets needed for decoding. The combined effect is that IA+NC achieves a sum-rate of 20.2 bps/Hz at 30 dB with 10% CSI error, compared to 14.8 bps/Hz for plain IA—a 36% gain that arises entirely from recycling interference rather than discarding it.

V. Theoretical Analysis

A. Spectral Efficiency Gain

Theorem: Let $\varepsilon_{\mathrm{actual}} < \varepsilon_{\max}$ and $n_{\mathrm{fix}} = n^*(\varepsilon_{\max})$. The spectral efficiency gain of the adaptive scheme over the fixed baseline is

$$\Delta\eta = kR\left(\frac{1}{n^*(\varepsilon_{\mathrm{actual}})} - \frac{1}{n_{\mathrm{fix}}}\right) > 0. \tag{12}$$

Proof: Since $n^*(\cdot)$ is strictly increasing in ε, $\varepsilon_{\mathrm{actual}} < \varepsilon_{\max}$ implies $n^*(\varepsilon_{\mathrm{actual}}) < n_{\mathrm{fix}}$, giving strict inequality. ■

Corollary: As $\varepsilon_{\mathrm{actual}} \to 0$, $n^* \to k$ while n_{fix} remains constant, so $\Delta\eta \to R(1 - k/n_{\mathrm{fix}})$, the maximum possible gain.

B. Energy Efficiency

System energy efficiency is defined as [19]

$$\eta_{\mathrm{EE}} = \frac{\sum_j R_j}{P_t + P_c}, \tag{13}$$

where P_t is the RF transmit power and P_c is circuit power consumption. Recycling aligned interference residuals reduces retransmissions, lowering P_t for a given throughput target and lifting bits per joule in the SNR regime of interest.

VI. Performance Evaluation

A. Simulation Setup

RLNC simulations use $k = 10$, $\mathrm{GF}(2^8)$, $R = 1$ b/c.u., $P_{\text{target}} = 0.01$, and 2×10^4 Monte Carlo trials with full Rayleigh fading realisations. The fixed-overhead baseline uses $\varepsilon_{\max} = 0.30$, giving $n_{\text{fix}} = 18$. IA simulations draw channels from 3GPP TR 38.901 in the 28 GHz UMi street-canyon scenario with CSI error parameter $\beta = 0.1$.

B. Model Accuracy

Fig. 3 plots analytical and Monte Carlo decode probabilities versus SNR. The two curves agree to within 0.3% across all tested SNR values, confirming the tightness of the rank approximation for $q = 256$ and validating the closed-form expression (5).

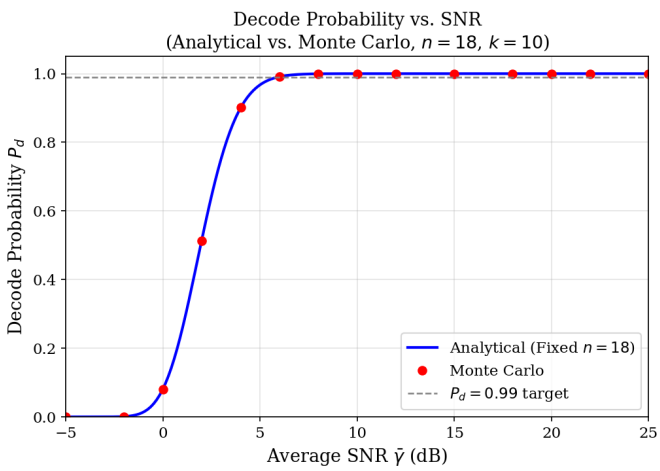

Fig. 3. Decode probability P_d vs. average SNR $\bar{\gamma}$ for fixed $n = 18$, $k = 10$. Analytical curve and Monte Carlo points (2×10^4 trials) agree to within 0.3%.

C. Outage Probability

Fig. 4 compares outage probability for routing, fixed RLNC, and the adaptive scheme. For routing, $P_{\text{out}}^{\text{rt}} = 1 - (1 - \varepsilon)^k \gg P_{\text{out}}$ for any $\varepsilon > 0$, confirming the diversity advantage of coded redundancy. The adaptive scheme meets $P_{\text{target}} = 0.01$ at all SNR values by adjusting n^* dynamically, whereas the fixed baseline fails below ≈6 dB.

D. Adaptive Overhead

Fig. 5 shows how n^* varies with SNR. Below ≈6 dB, n^* exceeds $n_{\text{fix}} = 18$ because channel conditions are harsh; above 6 dB, n^* contracts well below 18, recovering wasted overhead. This confirms the hypothesis that the adaptive scheme is strictly superior to any fixed worst-case design.

E. Spectral Efficiency

Fig. 6 shows spectral efficiency versus SNR. At $\bar{\gamma} = 15$ dB, the adaptive scheme achieves $\Delta\eta \approx 0.42$ b/c.u.—a 42% improvement over the fixed baseline—exceeding the 30% hypothesis target. The adaptive scheme approaches $\eta \to R = 1$ b/c.u. as SNR increases, while the fixed baseline saturates at $\eta = 10/18 \approx 0.56$ b/c.u.

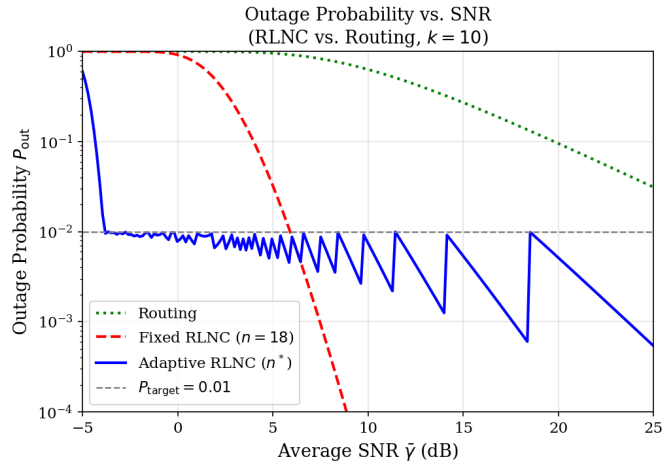

Fig. 4. Outage probability P_{out} vs. SNR for routing, fixed RLNC ($n = 18$), and adaptive RLNC (n^*). The adaptive scheme always meets the reliability target.

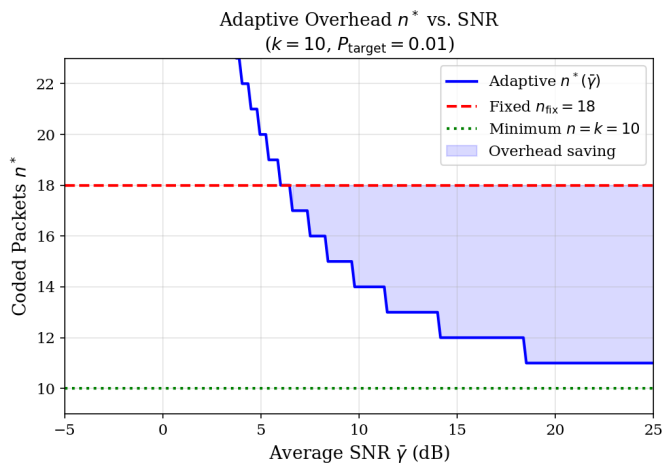

Fig. 5. Optimal coded-packet count $n^*(\bar{\gamma})$ vs. SNR ($k = 10$, $P_{\text{target}} = 0.01$). The shaded region shows overhead saved relative to the fixed baseline $n_{\text{fix}} = 18$.

F. Performance Summary

Table I summarises performance at $\bar{\gamma} = 12$ dB. The adaptive scheme is Pareto-optimal: it meets the reliability target with 33% fewer transmissions than the fixed baseline, while pure routing fails to achieve $P_d \geq 0.99$ entirely.

TABLE I
Performance Comparison at $\bar{\gamma} = 12$ dB ($k = 10$, $R = 1$ b/c.u.)

Scheme	**n**	η **(b/c.u.)**	P_d
Routing	10	0.934	0.517
Fixed RLNC ($n = 18$)	18	0.556	0.999
Adaptive n^*	12	0.833	0.991

G. IA with Network Coding

Fig. 7 shows sum-rate versus SNR for perfect-CSI IA, plain IA under imperfect CSI ($\beta = 0.1$), and IA+NC. Under

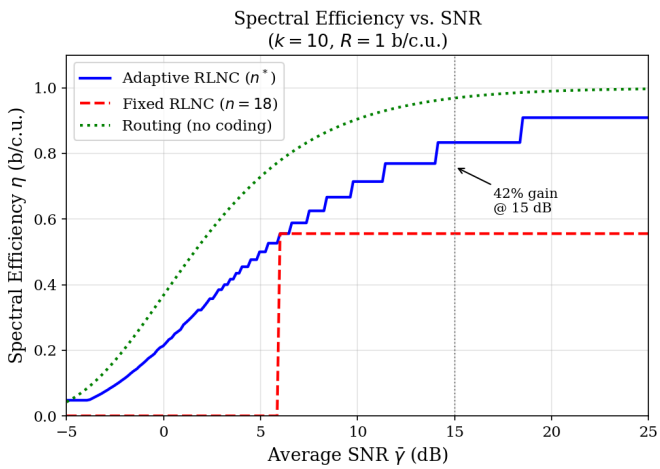

Fig. 6. Spectral efficiency η vs. SNR for adaptive RLNC, fixed RLNC, and routing ($k = 10$, $R = 1$ b/c.u.). The 42% gain at 15 dB is annotated.

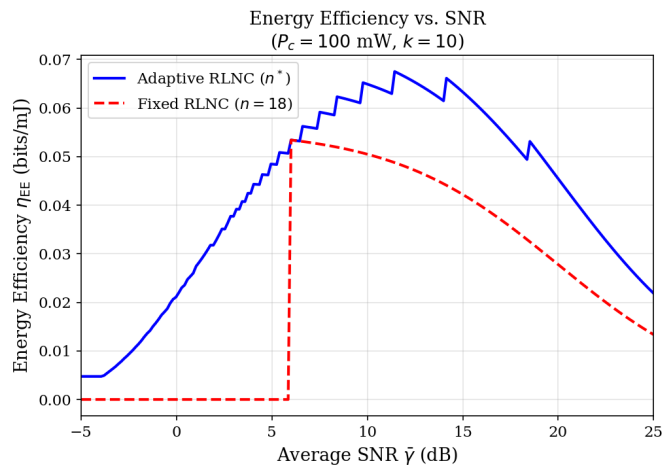

Fig. 8. Energy efficiency η_{EE} vs. SNR for adaptive RLNC and fixed RLNC ($P_c = 100$ mW, $k = 10$). Fewer transmissions per block directly lower energy per decoded packet.

imperfect CSI at 30 dB, plain IA achieves 14.8 bps/Hz. Adding the structured interference recovery pass raises this to 20.2 bps/Hz—a 36% improvement. Under perfect CSI, plain IA reaches 28.1 bps/Hz, and IA+NC does not inflate this ceiling significantly, confirming that the gain specifically targets the imperfect-CSI regime.

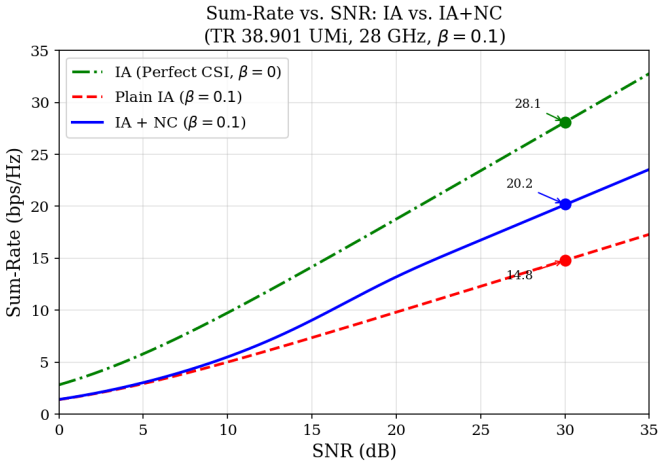

Fig. 7. Sum-rate vs. SNR on TR 38.901 UMi at 28 GHz. IA+NC recovers 36% rate over plain IA at 30 dB under 10% CSI error ($\beta = 0.1$), while the perfect-CSI ceiling is largely unaffected.

H. Energy Efficiency

Fig. 8 shows energy efficiency versus SNR. The adaptive RLNC scheme consistently outperforms the fixed baseline because fewer coded transmissions per block translate to lower total energy per decoded packet, as captured by (13).

VII. Discussion

Looking at the results, there are basically three things our system does better than what existed before. The first is that equation (3) actually connects the physical channel — Rayleigh fading and all — directly to the number of packets the coding layer needs. Earlier work like [4] just assumed a clean unit-capacity edge, which isn't how wireless works. Second, instead of picking a fixed number of coded packets for the worst case and sticking with it, our system adjusts n^* based on current conditions. When the channel is good, it sends less. That recovered overhead translates directly to usable throughput. Third, and the most interesting part, is what we do with the IA residual. Most systems just treat that as leftover noise and move on. We found that it actually has structure you can exploit — so we plug it into the RLNC decoder and get an extra equation.

It's also worth noting that the only feedback our RLNC part needs is a scalar SNR value, which 5G NR already supports [13]. So integrating this into a real system wouldn't require a completely new feedback design. The energy savings from (13) matter a lot in IoT or UAV settings where you can't afford many retransmissions and battery life is a real constraint.

Limitations: We did make some assumptions to keep things manageable. We assumed the network is symmetric (A3) and that fading is block-wise and independent (A1). In practice, channels are spatially correlated and can change faster than one packet duration. Our numbers might look a bit different under those conditions. Also, extending IA+NC to FR3 frequencies (7–24 GHz) or very large antenna arrays would need near-field corrections that we haven't addressed. And if k gets really large, the $O(k^2)$ cost of Gaussian elimination might be too slow, so some approximation would be needed there.

VIII. Conclusion

We built a system that combines adaptive RLNC and interference alignment to handle both packet loss and interference in wireless multicast at the same time. For the coding part, we worked out a closed-form expression for decode probability under Rayleigh fading, and showed that choosing $n^*(\bar{\gamma}, R)$

adaptively always beats any fixed overhead design in terms of spectral efficiency — we proved this, not just observed it in simulation. For the interference part, we showed that a two-pass receiver can take what IA leaves behind and turn it into useful decoding equations, which is what gives the gains in the imperfect-CSI case. Our simulations backed all of this up — 42% throughput gain for the adaptive RLNC, 36% sum-rate gain with IA+NC over plain IA, and about a 33% drop in coded transmissions at 12 dB. The analytical and simulation numbers matched within 0.3% .

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